Deep Learning – Transfer Learning

CIML Summer Institute 2026

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DEEP LEARNING OVERVIEW

- Neural Network Basics
  - Processing Unit
  - Activation Function
  - Loss Function
  - Neural Network Training
- Deep Learning Fundamentals
  - Deep Network Layers
  - DL Architectures
  - DL Libraries
- Transfer Learning
  - Transfer Learning Concepts
  - Transfer Learning Demo

Transfer Learning

- To overcome challenges of training model from scratch:
  - Insufficient data
  - Very long training time
- Use pre-trained model
  - Trained on another dataset
  - This serves as starting point for model
  - Then train model on current dataset for current task

Transfer Learning Approaches

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- Feature extraction
  - Remove classification layer from pre-trained model
  - Treat rest of network as feature extractor
  - Use features to train new classifier
    - “top model” or “classification head”
- Fine tuning
  - Tune weights in some layers of original model (along with weights of top model)
  - Train model for current task using new dataset

CNNs for Transfer Learning

- Popular architectures
  - AlexNet
  - GoogLeNet
  - VGGNet
  - ResNet
- All winners of ILSVRC
  - ImageNet Large Scale Visual Recognition Challenge
  - Annual competition on vision tasks on ImageNet data

ImageNet

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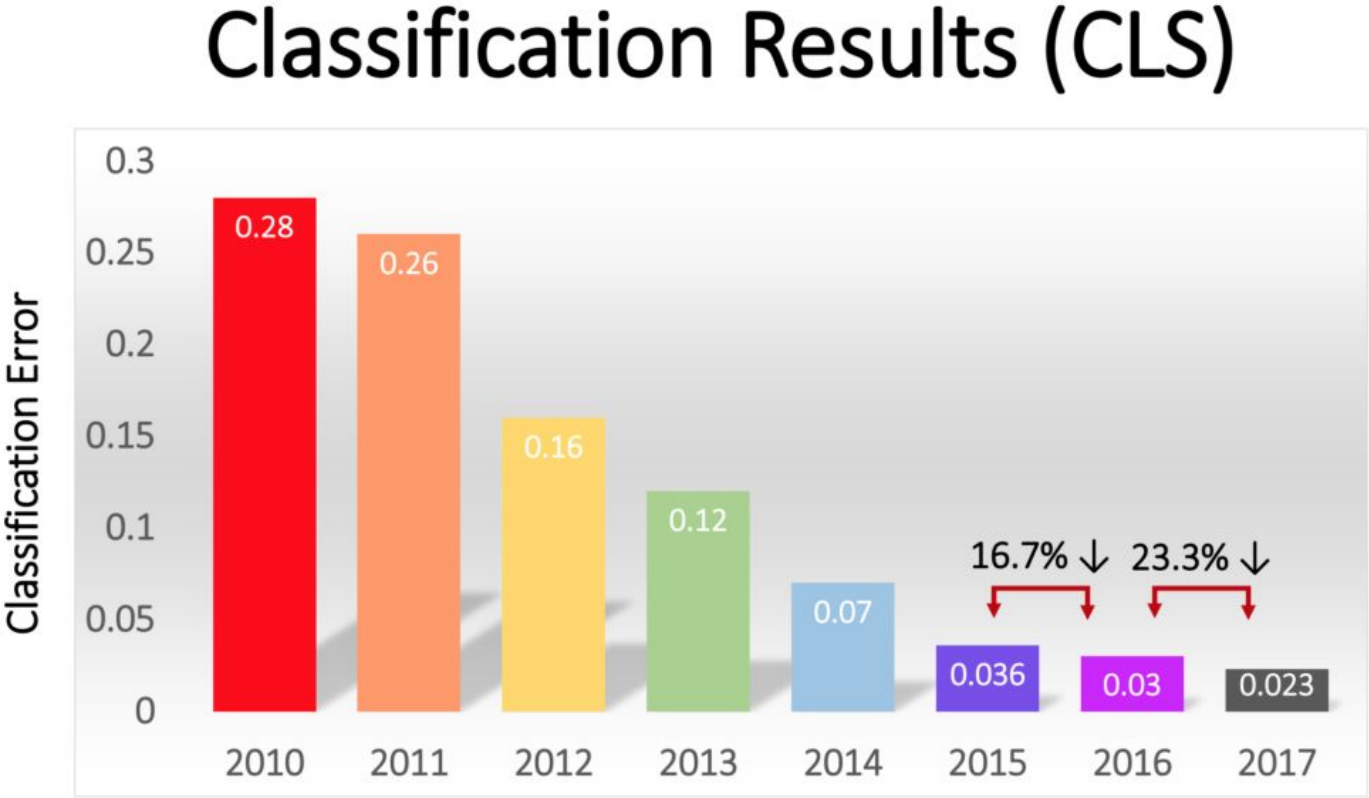
- Database
  - Developed for computer vision research
  - ~14,000,000 images hand-annotated
  - ~22,000 categories
- ILSVRC History
  - Started in 2010
  - Image classification task: 1,000 object categories
  - Image classification error rate
    - 2010: 28.20%   (conventional image processing techniques)
    - 2012: 15.30% (AlexNet)
    - 2015:   3.57% (ResNet; better than human performance)
    - 2016:   2.99% (16.7% error reduction)
    - 2017:   2.25% (23.3% error reduction)

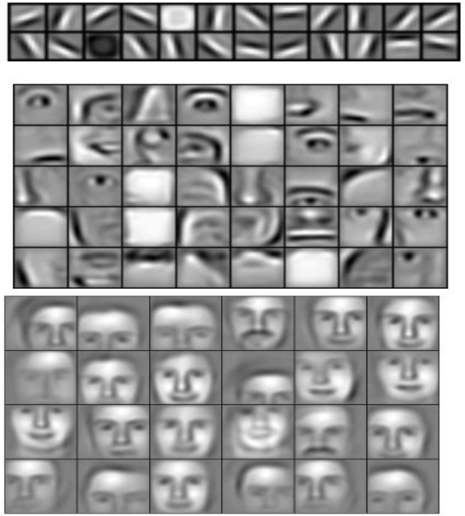
Results on ImageNet Classification



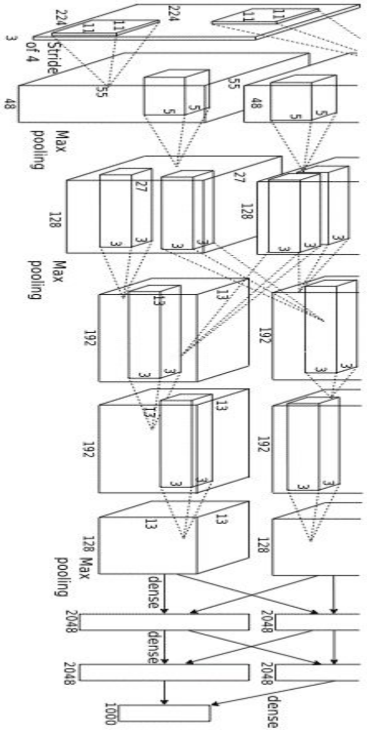
Transfer Learning

Input

Learned hierarchy

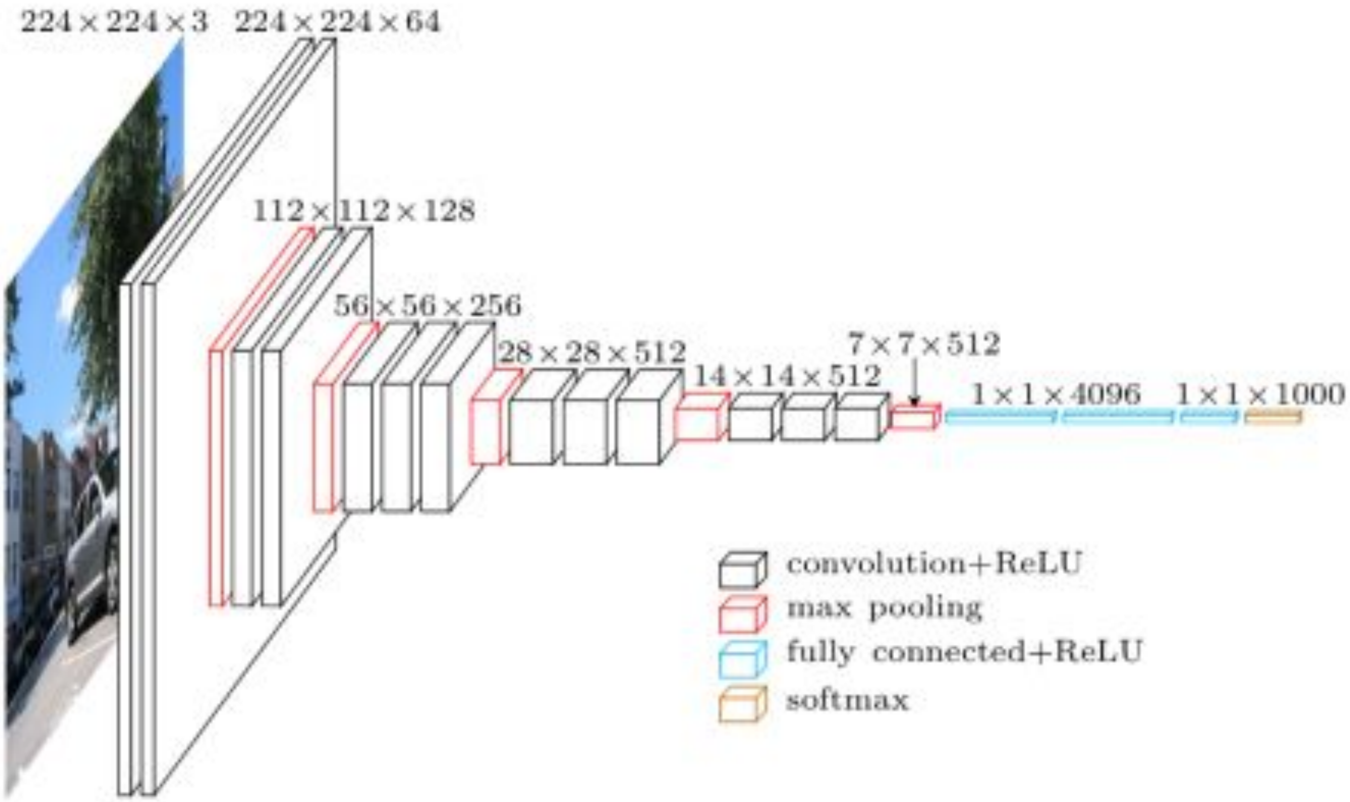


Output



Lee et al. 'Convolutional Deep Belief Networks for Scalable Unsupervised Learning of Hierarchical Representations' ICML 2009

Pre-Trained Model



https://www.cs.toronto.edu/~frossard/post/vgg16/

Transfer Learning - Feature Extraction

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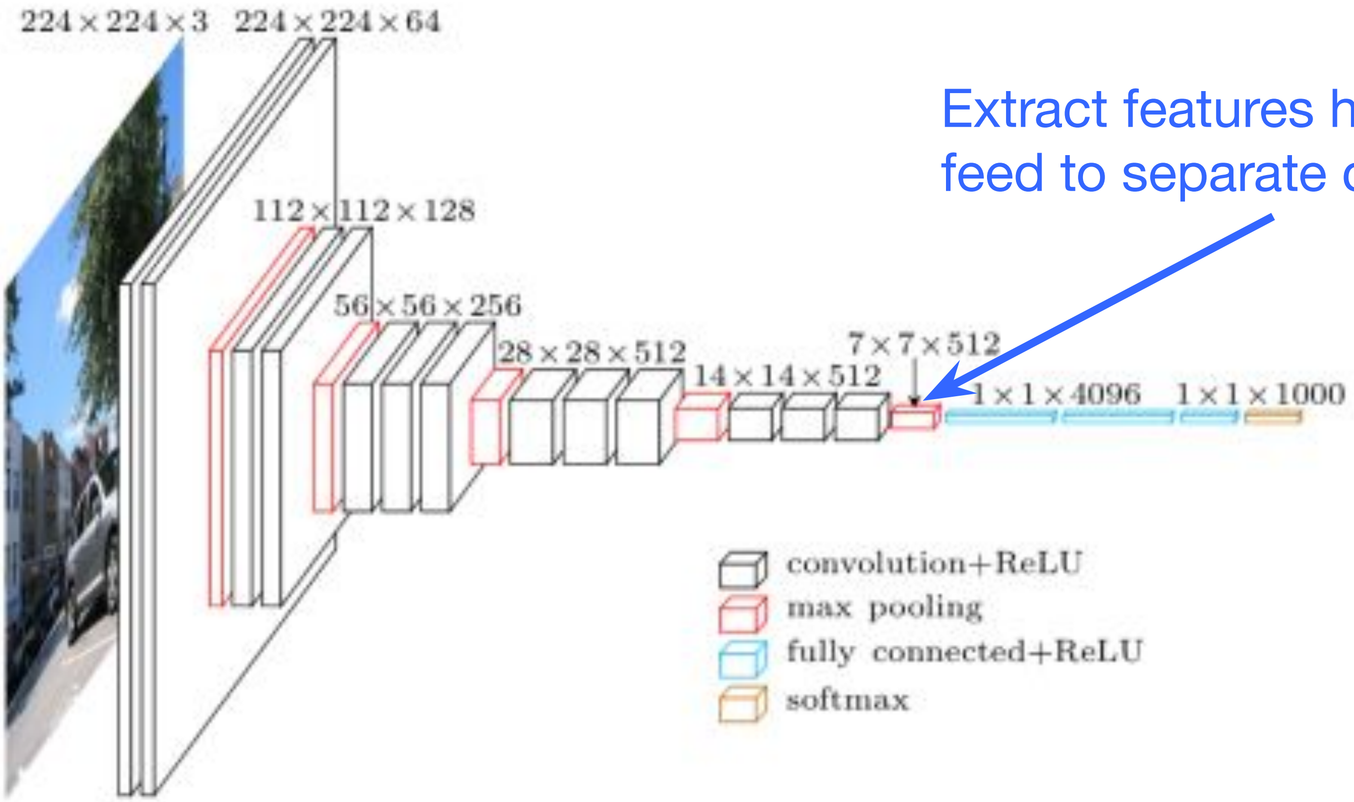
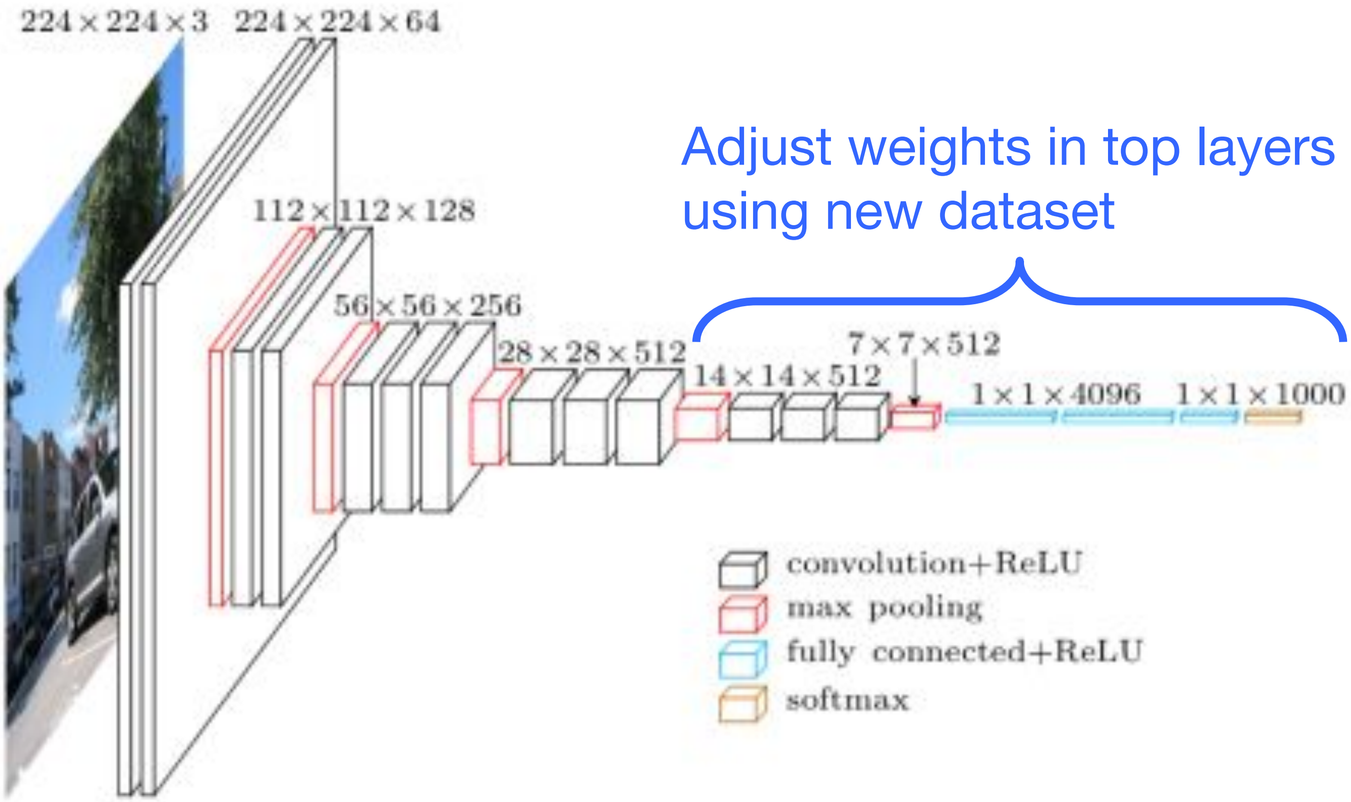


Extract features here and feed to separate classifier

https://www.cs.toronto.edu/~frossard/post/vgg16/

Transfer Learning - Finetuning



https://www.cs.toronto.edu/~frossard/post/vgg16/

Practical Tips for Transfer Learning

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- Learning rate
  - Use very small learning rate for fine tuning. Don't want to destroy what was already learned.
- Start with properly trained weights
  - Train top-level classifier first, then fine tune lower layers.
  - Top model with random weights may have negative effects on when fine tuning weights in pre-trained model
- Data augmentation
  - Simple ways to slightly alter images
    - Horizontal/vertical flips, random crops, translations, rotations, etc.
  - Use to artificially expand your dataset

Transfer Learning Hands-On

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- Data
  - Cats and dogs images from Kaggle
- Exercises
  - Feature extraction
    - Use pre-trained CNN to extract features from images
    - Train neural network to classify cats/dogs using extracted features
  - Fine tune
    - Adjust weights of last few layers of pre-trained CNN and top classifier model through training

Data

- Subset of Dogs Vs. Cats dataset from Kaggle
  - https://www.kaggle.com/c/dogs-vs-cats
- Train
  - 1000 cats + 1000 dogs
- Validation
  - 200 cats + 200 dogs
- Test
  - 200 cats + 200 dogs

Transfer Learning - Feature Extraction

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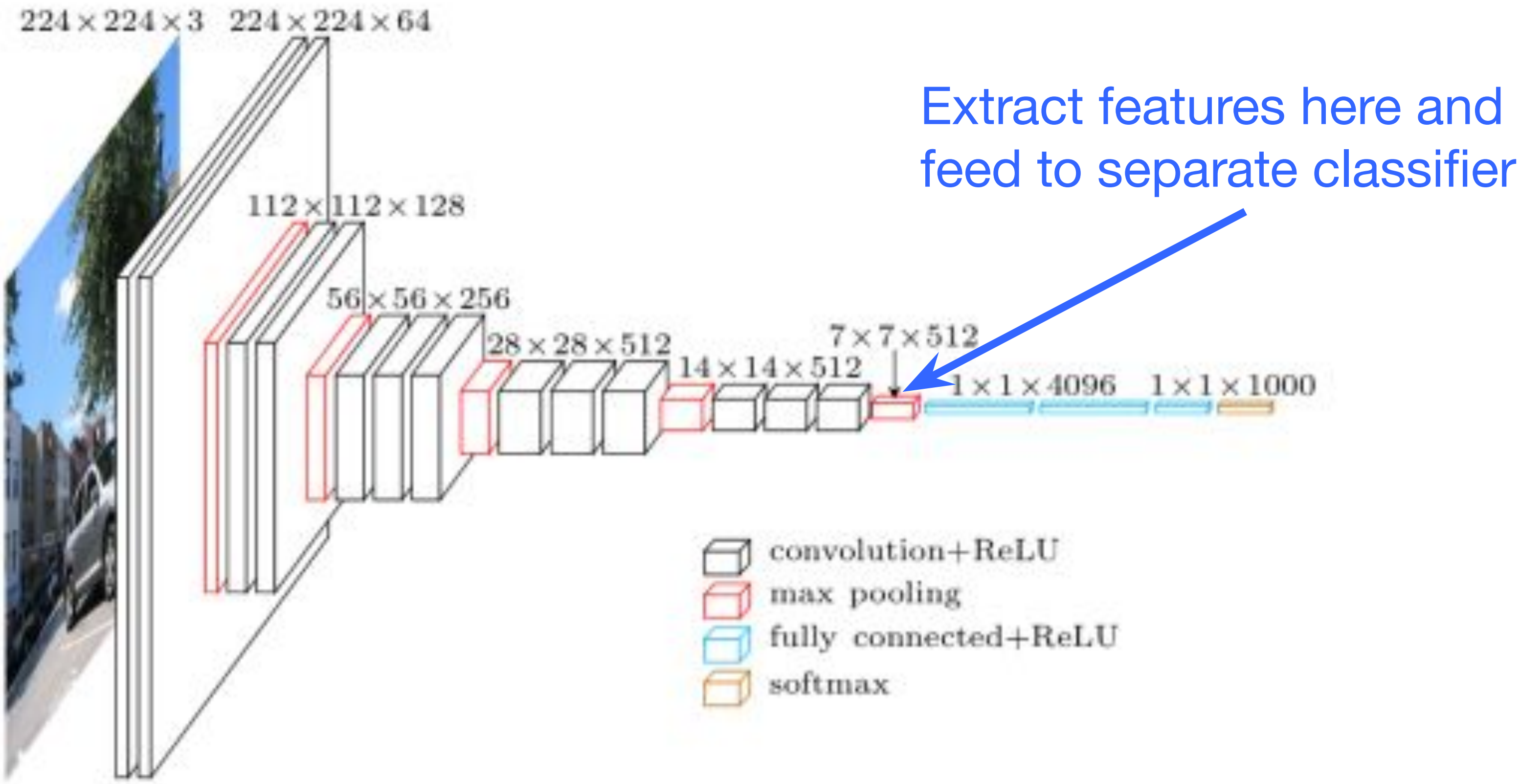

https://www.cs.toronto.edu/~frossard/post/vgg16/

Feature Extraction Overview

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- Code
  - feature_extraction_ptl.ipynb
- Data
  - Set image dimensions & location
  - Read images from folder in batches
- Model
  - Load model pre-trained on ImageNet data
  - Freeze weights in pre-trained model to use as feature extractor
  - Add top model to classify cats vs dogs
  - Model = Pre-trained base model + top model classifier
- Train model
  - Use training data to adjust top model weights
- Evaluate model
  - Calculate accuracy, etc.
  - Perform inference on test images

Server Setup on Expanse

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- Login to Expanse through terminal window
  - Open terminal window on local machine and type in:
    ```
    ssh login.expanse.sdsc.edu -l <username>
    ```
- Pull latest from repo
  - git pull
  - URL: https://github.com/ciml-org/ciml-summer-institute-2026
- In terminal window
  - `jupyter-nairr-gpu-shared-pytorch`
    - Alias for:
      ```
      galyleo launch --account ${CIML26_ACCOUNT} --reservation
      $CIML26_RES_GPU} --partition nairr-gpu-shared --qos
      ${CIML26_QOS_GPU} --cpus 4 --memory 92 --gpus 1 --time-limit
      04:00:00 --sif ${CIML26_DATA_DIR}/ptl-cuda-12-1.sif --bind
      /expanse,/scratch,/cm --nv --quiet
      ```
  - Copy and paste URL in local web browser
- To check queue
  - `squeue -u $USER`

Data

- In terminal window in Jupyter Lab, do the following
- Get counts of images
  - `ls`
  - `ls -l $CIML26_DATA_DIR/catsVsDogs/train/cats/* | wc -l`
  - `ls -l $CIML26_DATA_DIR/catsVsDogs/train/dogs/* | wc -l`
  - `ls -l $CIML26_DATA_DIR/catsVsDogs/val/cats/*  | wc -l`
  - `ls -l $CIML26_DATA_DIR/catsVsDogs/val/dogs/* | wc -l`
  - `ls -l $CIML26_DATA_DIR/catsVsDogs/test/cats/*  | wc -l`
  - `ls -l $CIML26_DATA_DIR/catsVsDogs/test/dogs/* | wc -l`

Transfer Learning - Finetuning

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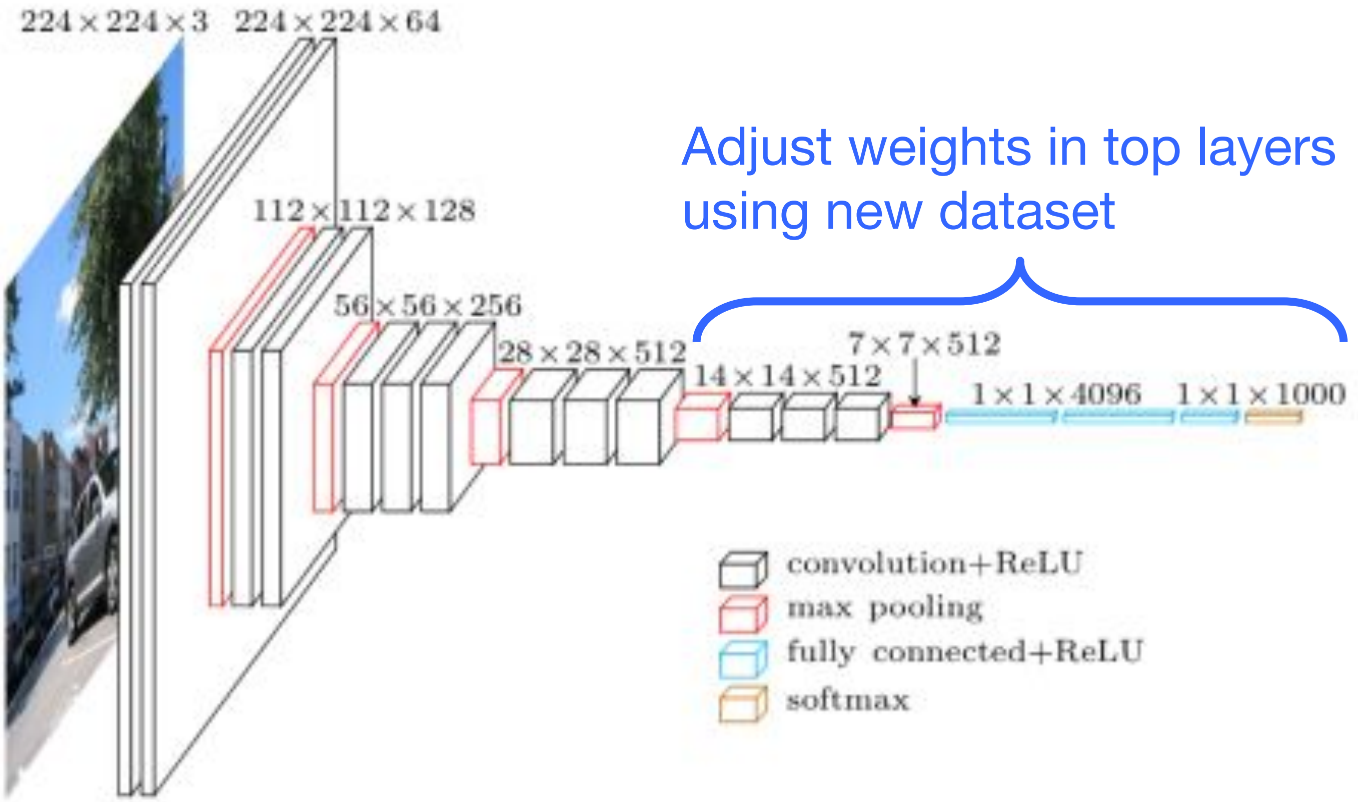

Adjust weights in top layers using new dataset

https://www.cs.toronto.edu/~frossard/post/vgg16/

Fine Tune Overview

- Code
  - finetune_ptl.ipynb
- Data
  - Set image dimensions & location
  - Read images from folder in batches
- Model
  - Load trained model from feature extraction code
  - Weights in last few convolutional blocks and top model will be adjusted during training
  - All other weights in pre-trained model are frozen
- Train model
  - Use training data to adjust top model weights
  - Use validation data to determine when to stop training
- Evaluate model
  - Calculate accuracy, etc.
  - Perform inference on test images

Early Stopping

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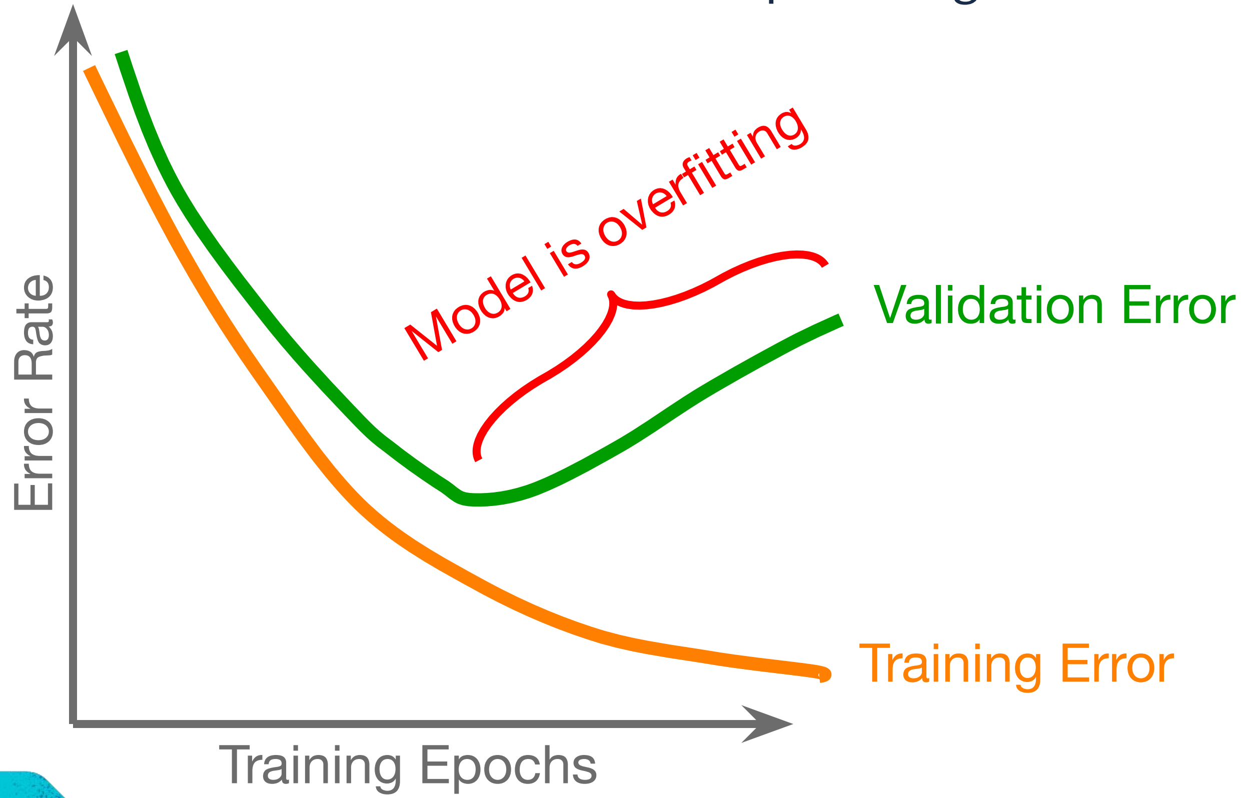
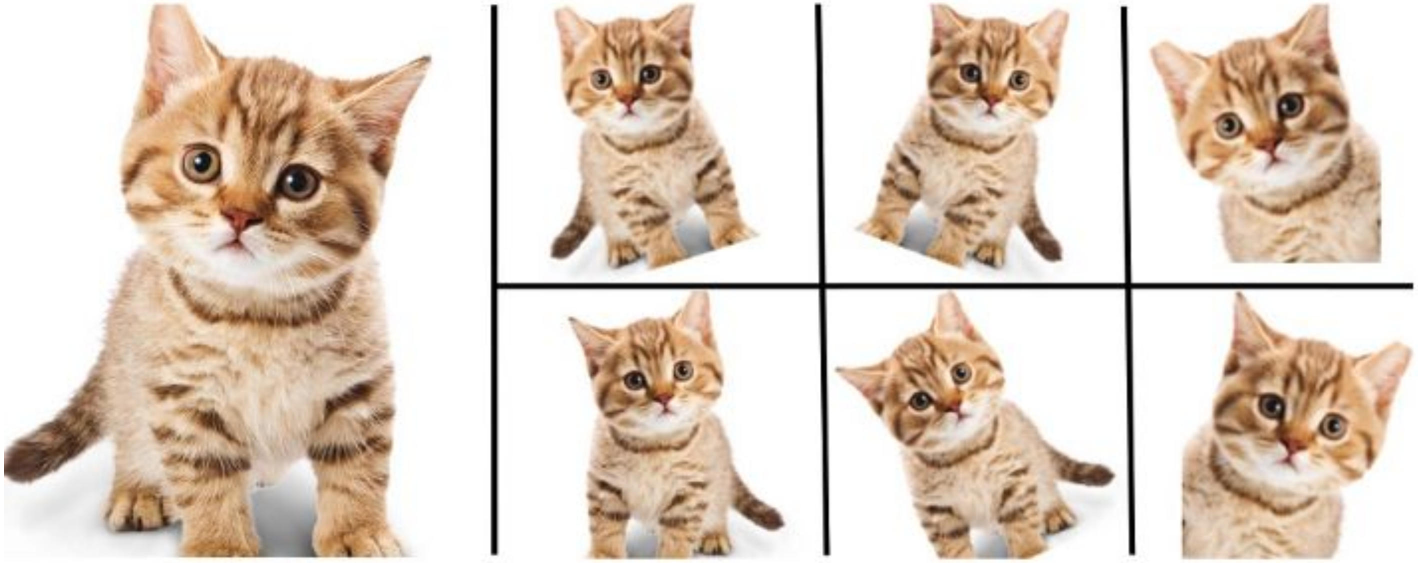
Using validation data to determine when to stop training to avoid overfitting



Data Augmentation



Add variability to your dataset

https://nanonets.com/blog/data-augmentation-how-to-use-deep-learning-when-you-have-limited-data-part-2/

RESOURCES

- Transfer Learning
  - http://cs231n.github.io/transfer-learning/
- ImageNet
  - http://www.image-net.org/